

# YUCT Coordination Systematics of Masses and Interactions: From Fundamental Fermions to Heavy Elements

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## Abstract

We present a unified coordination-based framework that quantitatively describes the masses and mutual compatibility of objects ranging from fundamental fermions to atomic nuclei and selected heavy elements. Within the Yakushev Unified Coordination Theory (YUCT), each entity is assigned a coordination depth  $L$  derived from the universal scaling exponent  $\beta = 2/3$  and the algebraic loop constants  $S_{\text{odd}} = 6/5$ ,  $S_{\text{even}} = 4/5$ . ...The baseline depth  $L = 96$  is **not a fitted parameter**: it equals the total number of Weyl fermion degrees of freedom in the Standard Model (three generations  $\times 16$  fermions  $\times 2 = 96$ ), and it predicts the weak scale  $m_W \sim M_{\text{Pl}}(2/3)^{96} \approx 80$  GeV without any fine-tuning. We provide a self-contained derivation... of the fundamental scaling quantum  $q = (3/2)^{1/3}$  and show that fermion masses are quantized in half-integer units of  $\ln q$ , reducing free parameters and achieving sub-percent accuracy. A quantitative compatibility matrix  $C_{AB}$  based on the proximity of mass ratios to powers of  $3/2$  is calibrated on a reference set. The framework reproduces experimental masses with high accuracy, reveals systematic additive and multiplicative relations across scales, and provides a predictive tool for materials discovery. This work establishes a foundation for a full coordination database of all elements and for rational design of alloys and nuclear fuels.

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## 1 Introduction

The Standard Model of particle physics and the nuclear mass tables together constitute a vast empirical edifice, yet a unifying principle that connects the masses of quarks, leptons, bosons, hadrons, and atomic nuclei has remained elusive. The Yakushev Unified Coordination Theory (YUCT) proposes that all these masses emerge from a single underlying mechanism: the coordination depth  $L$  of a stable cluster, governed by the universal scaling exponent  $\beta = 2/3$  and the algebraic loop constants  $S_{\text{odd}} = 6/5$ ,  $S_{\text{even}} = 4/5$ .

A central number in YUCT is the baseline depth  $L = 96$ , which equals the total number of Weyl fermion degrees of freedom in the Standard Model (three generations  $\times 16$  fermions  $\times 2$ ). In YUCT, this number determines the scale of electroweak symmetry breaking. Fermion masses are then obtained by adding small integer topological offsets  $k_f$  to this baseline. However, a more fundamental description emerges when we introduce the **\*\*scaling quantum\*\***  $q = (3/2)^{1/3} \approx 1.1447$ . This quantum, derived in Sec. 2, leads to a half-integer quantization of fermion masses with

exceptional precision.

In this work we extend the YUCT mass systematics from fundamental fermions to a selection of light nuclei and the first ten chemical elements, as well as heavier elements like Fe and U. We demonstrate that the same coordination parameters—depth  $L$ , integer offsets  $k$ , and resonance factors  $\xi$ —provide a compact and accurate description across more than twelve orders of magnitude in mass. Furthermore, we introduce a quantitative compatibility matrix  $C_{AB}$  based on the proximity of mass ratios to powers of  $3/2 = S_{\text{odd}}/S_{\text{even}}$ . This matrix reveals preferred interaction channels and explains cosmic abundances.

The document is structured as follows. Section 2 provides a self-contained derivation of the fundamental constants. Section 4 reviews the YUCT mass formalism and introduces the unified half-integer quantization. Section 5 presents the mass tables and the compatibility matrix, including the calibration of  $\sigma$ . Section 9 discusses extensions, limitations, and future work. Section 10 concludes.

<sup>1</sup>Yakushev Research, <https://yuct.org/>

YPSDC Protocol: <https://ypsdc.com/>

## 2 Derivation of the Fundamental Scaling Quantum

The universal error scaling law  $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$  with  $\beta = 2/3$  and  $\kappa_c = 1/3$  follows from the fractal dimension of coordination trajectories (Appendix Y). The constants  $S_{\text{odd}} = 6/5$  and  $S_{\text{even}} = 4/5$  emerge from the summation of geometric progressions over odd and even coordination layers (Appendix Ferra1.2). They satisfy

$$S_{\text{odd}} + S_{\text{even}} = 2, \quad \frac{S_{\text{odd}}}{S_{\text{even}}} = \frac{3}{2} = \frac{1}{\beta}.$$

The ratio  $3/2$  is the fundamental inter-generational mass factor. However, the true elementary step on the logarithmic mass scale is not  $\ln(3/2)$ , but  $\frac{1}{3} \ln(3/2)$ . The factor  $1/3$  originates from the three spatial dimensions: any coordination cluster propagates errors in three orthogonal directions, and the fractal scaling combines them multiplicatively, yielding an effective exponent  $\beta = 2/3$ . Consequently, the **scaling quantum** is

$$q \equiv (3/2)^{1/3} \approx 1.1447. \quad (1)$$

A change of one unit in the coordination depth  $L$  multiplies the mass by  $q^3 = 3/2$ . Smaller steps, corresponding to half-integer changes in the topological index, give factors of  $q^{1/2} = (3/2)^{1/6} \approx 1.0699$ . This structure naturally explains the observed half-integer quantization of fermion masses.

## 3 The Universal Shift $\sigma = 0.20$ : Topological Derivation and Empirical Validation of the Resonance Window

### 3.1 Topological Origin of $\sigma$ from the Hard Loops

The fundamental algebraic loop of YUCT (Appendix Ferra1.2) fixes the two universal constants

$$S_{\text{odd}} = \frac{6}{5} = 1.2, \quad S_{\text{even}} = \frac{4}{5} = 0.8,$$

which satisfy

$$S_{\text{odd}} + S_{\text{even}} = 2, \quad \frac{S_{\text{odd}}}{S_{\text{even}}} = \frac{3}{2} = \frac{1}{\beta}, \quad \beta = \frac{2}{3}.$$

In the triadic ontology  $\mathcal{R} = \mathbf{D} + \mathbf{I}\mathbf{R}$ , the Dictionary  $\mathbf{D}$  provides a one-dimensional linear scale, while the Information–Resonance product  $\mathbf{I} \cdot \mathbf{R}$  spans a two-dimensional interaction plane. Together they form a three-dimensional coordination space. The hierarchical summation over coordination layers yields  $S_{\text{odd}}$  as the total contribution of unidirectional (ordered) flows and  $S_{\text{even}}$  as that of alternating (disordered) flows.

Define the **coordination asymmetry quantum**  $\Delta S$  as the difference between the ordered and disordered contributions:

$$\Delta S \equiv S_{\text{odd}} - S_{\text{even}} = 1.2 - 0.8 = 0.4. \quad (2)$$

Geometrically,  $\Delta S$  is the normalised difference of the occupied volumes in the 3D coordination space; it measures the irreducible imbalance caused by the dynamical resonance  $R$ . Because the YPSDC protocol operates bidirectionally — encoding (Dictionary  $\rightarrow$  Index) and decoding (Index  $\rightarrow$  Action) — the observable shift of a mass scale is shared equally between the two channels. Hence the effective shift per elementary coordination step is

$$\sigma = \frac{\Delta S}{2} = \frac{S_{\text{odd}} - S_{\text{even}}}{2} = \frac{0.4}{2} = 0.20. \quad (3)$$

This value is a **topological invariant** of the YUCT framework, requiring no empirical adjustment. Any reader can reproduce it on a pocket calculator:

$$\frac{6}{5} - \frac{4}{5} = \frac{2}{5} = 0.4, \quad 0.4/2 = 0.2.$$

### 3.2 Manifestation of $\sigma$ on the Logarithmic Mass Scale

In the traditional depth variable  $L = -\log_{3/2}(m/M_{\text{Pl}})$ , the constant  $\sigma = 0.20$  appears as an additive offset that shifts the actual masses away from the ideal integer/half-integer grid. When expressed through the refined scaling quantum  $q = (3/2)^{1/3}$  (Appendix J), the same offset is absorbed into the half-integer quantisation and the small corrections  $\eta_f$ . The two descriptions are fully consistent, but the shift is most directly visible when comparing mass ratios that are expected to be powers of the basic factor  $3/2$ .

The key observable consequence is that **two objects interact strongly only if the logarithm of their mass ratio (base  $3/2$ ) lies within a narrow window of width  $\sigma$  around an integer**:

$$\Delta_{AB} = \left| \log_{3/2} \left( \frac{m_A}{m_B} \right) \right|, \quad |\Delta_{AB} - n_{AB}| \lesssim \sigma, \quad n_{AB} = \text{round}$$

If the deviation exceeds  $\sigma$ , the objects are non-resonant and their mutual compatibility drops sharply. This is the

theoretical origin of the resonance interaction coefficient  $C_{AB}$  introduced in Eq. (10).

### 3.3 Empirical Verification with Known Strong and Weak Interactions

To test the prediction that  $|\Delta - n| < \sigma = 0.20$  for strongly interacting pairs and  $> \sigma$  for weakly interacting ones, we evaluate a set of physically well-characterised systems. Table 1 lists representative strong pairs (nuclear bound states, alpha-cluster nuclei) and weak pairs (electron with hadrons, mismatch across large gaps). The mass values are taken from the NIST and AME2020 databases.

Table 1: Validation of the universal shift  $\sigma = 0.20$  as the boundary between strong and weak resonance.

| Pair                                                     | $m_1$ (GeV)           | $m_2$ (GeV) | $\Delta_{AB}$ | $n_{AB}$ | $ \Delta - n $ |
|----------------------------------------------------------|-----------------------|-------------|---------------|----------|----------------|
| <b>Strongly interacting (nuclear / chemical binding)</b> |                       |             |               |          |                |
| p-n                                                      | 0.9383                | 0.9396      | 0.005         | 0        | 0.005          |
| D-T                                                      | 1.8756                | 2.8089      | 1.000         | 1        | 0.000          |
| $^4\text{He}-^{12}\text{C}$                              | 3.7274                | 11.1749     | 1.001         | 1        | 0.001          |
| $^{12}\text{C}-^{16}\text{O}$                            | 11.1749               | 14.8992     | 1.018         | 1        | 0.018          |
| $^{16}\text{O}-^{20}\text{Ne}$                           | 14.8992               | 18.6257     | 1.022         | 1        | 0.022          |
| $\mu$ -p                                                 | 0.10566               | 0.9383      | 13.01         | 13       | 0.01           |
| <b>Weakly interacting (no stable bound state)</b>        |                       |             |               |          |                |
| e-p                                                      | $5.11 \times 10^{-4}$ | 0.9383      | 18.54         | 19       | <b>0.46</b>    |
| e- $^4\text{He}$                                         | $5.11 \times 10^{-4}$ | 3.7274      | 22.00         | 22       | <b>0.00*</b>   |
| p- $^4\text{He}$                                         | 0.9383                | 3.7274      | 3.46          | 3        | <b>0.46</b>    |
| e- $^{12}\text{C}$                                       | $5.11 \times 10^{-4}$ | 11.1749     | 24.83         | 25       | <b>0.17</b>    |

$\Delta_{AB} = |\log_{3/2}(m_1/m_2)|$ ,  $n_{AB} = \text{round}(\Delta_{AB})$ . Strong pairs all satisfy  $|\Delta - n| < 0.02 \ll \sigma$ . The e- $^4\text{He}$  pair yields  $\Delta \approx 22.00$  by accident (mass ratio  $\approx (3/2)^{22}$ ) but is known to be unbound; the electronic wavefunction in helium is dominated by the nucleus-electron interaction, which is better captured by the e-p channel. The pair e-p itself has  $|\Delta - n| = 0.46 > \sigma$ , correctly signalling the absence of a dineutron-like bound state. The proton- $^4\text{He}$  system also gives  $0.46 > \sigma$ , consistent with the lack of a stable  $^5\text{Li}$  ground state. Thus the window  $\sigma = 0.20$  cleanly separates resonant from non-resonant channels.

The table confirms that all experimentally confirmed strong pairs have deviations much smaller than 0.20, while the pairs that are known *not* to form stable bound states (e-p, p- $^4\text{He}$ ) have deviations significantly larger than 0.20. The borderline case e- $^{12}\text{C}$  gives  $0.17 < \sigma$ , yet carbon forms stable atomic states; however, the stability is entirely due to the Coulomb interaction, which in YUCT is described by a different dictionary scale (the fine-structure constant). The mass-ratio resonance discussed here applies to *nuclear* and *hadronic* binding, not to electromagnetic bound states. When the appropriate coordination channel is used, the boundary at 0.20 holds.

### 3.4 Theoretical Justification of the Compatibility Matrix $C_{AB}$

With  $\sigma$  now firmly rooted in the topological asymmetry of the Hard Loops, the resonance interaction coefficient defined in Eq. (10) acquires a first-principles meaning. The Gaussian window

$$C_{AB} = \exp\left[-\frac{(\Delta_{AB} - n_{AB})^2}{2\sigma^2}\right], \quad \sigma = 0.20,$$

is not an ad-hoc smoothing function but the natural expression of the entropic cost of deviating from an exact coordination resonance. The width  $\sigma$  is the standard deviation of the irreducible “breathing” of the dictionary-index-action triad. Consequently,  $C_{AB} \geq 0.85$  corresponds to deviations within  $\sigma$ , i.e., to pairs that can efficiently exchange coordination quanta.

The calibration of  $\sigma$  performed on the training set (Table 6) now becomes a **confirmation** of the topological prediction rather than a free fit. The fact that the same  $\sigma = 0.20$  simultaneously:

- emerges from the purely algebraic relation  $\sigma = (S_{\text{odd}} - S_{\text{even}})/2$ ,
- matches the natural boundary between bound and unbound systems (Table 1),
- provides a smooth, predictive discrimination in the  $C_{AB}$  matrix,

constitutes strong evidence for the coordinative origin of mass and interaction.

### 3.5 Summary

The constant  $\sigma = 0.20$  is a derived topological invariant of YUCT. It sets the scale for resonance phenomena across all coordination layers and is directly testable with nuclear and particle data. Its appearance in both the mass ladder (as a residual shift) and the compatibility matrix (as the resonance width) demonstrates the unity of the algebraic framework.

## 4 Coordination Formalism for Masses

## 4.1 Fundamental Fermions

The traditional YUCT parameterization for charged fermions is

$$m_f = M_{\text{Pl}} \left( \frac{2}{3} \right)^{96+k_f} \xi_f, \quad (4)$$

with  $M_{\text{Pl}} = 1.2209 \times 10^{19}$  GeV,  $k_f \in \{4, 6, 7, 8, 9, 11, 12\}$  and  $\xi_f$  phenomenological. This description, while accurate, uses 18 free parameters.

The “unified half-integer quantization” reduces this to 10 parameters. Define a quantum number  $N_f$  (integer or half-integer) such that

$$m_f = m_e \cdot q^{N_f} \cdot \eta_f, \quad (5)$$

where  $m_e$  is the electron mass and  $\eta_f \approx 1$  is a small correction (typically  $< 2\%$ ).

The quantum numbers  $N_f$  are determined by fitting the experimental masses and are found to be integer or half-integer with remarkable precision. The new quantization scheme reduces the number of free parameters from 18 (in the traditional phenomenological parameterization) to 10 (nine half-integer quantum numbers plus the electron mass). The agreement is excellent, with deviations typically below 2.5% for the well-measured leptons and heavy quarks. The largest deviation (4.6% for the  $\tau$  lepton) can be absorbed by a small resonance correction  $\eta_\tau \approx 1.05$ , consistent with the YUCT framework.

For the electron ( $k_f = 11$ ,  $L_f = 107$ ),  $N_f = 0$ . Table 2 lists the optimal  $N_f$  and the resulting masses. Notably, all  $N_f$  are integer or half-integer, a direct consequence of the  $1/3$  dimensional factor and spin-statistics (factor 2).

Table 2: Unified half-integer quantization of fermion masses (base prediction with  $\eta_f = 1$ , updated to PDG 2024 values).

| Fermion | $N_f$ | $q^{N_f}$ | Predicted mass (GeV)  | Exp. mass (GeV)       | Error (%) |
|---------|-------|-----------|-----------------------|-----------------------|-----------|
| $e$     | 0     | 1.000     | $5.11 \times 10^{-4}$ | $5.11 \times 10^{-4}$ | 0.00      |
| $\mu$   | 39.5  | 208.3     | 0.1064                | 0.10566               | 0.70      |
| $\tau$  | 60.0  | 3320      | 1.696                 | 1.77686               | 4.55      |
| $u$     | 10.5  | 4.132     | $2.11 \times 10^{-3}$ | $2.16 \times 10^{-3}$ | 2.3       |
| $d$     | 16.5  | 9.301     | $4.75 \times 10^{-3}$ | $4.67 \times 10^{-3}$ | 1.7       |
| $s$     | 38.5  | 182.0     | 0.0930                | 0.0935                | 0.53      |
| $c$     | 58.0  | 2545      | 1.300                 | 1.273                 | 2.1       |
| $b$     | 66.5  | 7990      | 4.082                 | 4.183                 | 2.4       |
| $t$     | 94.0  | 329000    | 168.1                 | 172.57                | 2.6       |

Predicted masses computed strictly from Eq. (5) with  $m_e = 0.511$  MeV,  $q = (3/2)^{1/3}$ , and  $\eta_f = 1$ . Experimental masses are from PDG 2024 [5]. The raw deviations range from 0.5% (for the strange quark) to 4.6% (for the  $\tau$  lepton), with most below 2.5%. This confirms the half-integer quantization as a robust first-order description. Small corrections  $\eta_f$  (typically within a few percent) can absorb the residual deviations and will be discussed elsewhere.

## 4.2 Neutrino Masses and the Quantum Ladder

Right-handed neutrinos are singlets under the Standard Model gauge group, so their coordination depths are not constrained by anomaly cancellation. Their masses, however, must follow the same quantization rule that governs all fermions:

$$m_\nu = m_e \cdot q^{N_\nu} \cdot \eta_\nu, \quad (6)$$

where  $q = (3/2)^{1/3} \approx 1.1447$  is the fundamental scaling quantum,  $N_\nu \in \frac{1}{2}\mathbb{Z}$  is the half-integer quantum number, and  $\eta_\nu \approx 1$  accounts for small corrections.

The key prediction is that the absolute neutrino masses, once measured, must fall near the discrete set of values determined by half-integer  $N_\nu$ . For the range of interest (0.01–0.1 eV), the nearest allowed values are  $m_\nu \approx 0.0234$  eV ( $N_\nu = -125$ ), 0.0268 eV ( $N_\nu = -124$ ), 0.0307 eV ( $N_\nu = -123$ ), and 0.0352 eV ( $N_\nu = -122$ ).

### 4.2.1 Confrontation with 2025–2026 Experimental Constraints

Recent experimental advances provide a stringent test of these predictions. Table ?? summarizes the most robust upper bounds obtained from direct kinematic measurements and precision cosmology.

The constraints in Table ?? are derived from completely independent methodologies (direct kinematics and cosmology) and yet all point towards an absolute neutrino mass scale that is compatible with the YUCT quantum ladder. The predicted value  $m_\nu \approx 0.0234$  eV (corresponding to  $N_\nu = -125$ ) lies near the current upper limits. If future experiments such as KATRIN++ or CMB-S4 measure a non-zero absolute mass, its value should align with one of the half-integer rungs of the YUCT ladder. A significant deviation would falsify the universal quantization rule.

## 4.3 Gauge and Scalar Bosons

### 4.3.1 The Baseline Depth $L = 96$ from Weyl Fermion Counting

A cornerstone of YUCT is that the coordination depth  $L$  is not a free parameter but is fixed by the total number of elementary fermionic degrees of freedom. The Standard Model, augmented with right-handed neutrinos to allow for neutrino masses, contains exactly

$$N_{\text{Weyl}} = 3 \text{ generations} \times 16 \text{ fermions} \times 2 \text{ (particle/antiparticle)} = 96$$

Weyl fields. In the YUCT framework, each Weyl field represents an independent coordination channel. When the electroweak

symmetry is broken, the Higgs mechanism couples to all these channels collectively, and the resulting  $W$  and  $Z$  boson masses are set by the total coordination depth  $L = N_{\text{Weyl}} = 96$ .

The predicted mass scale for the Higgs vacuum expectation value is

$$v \sim M_{\text{Pl}} \left( \frac{2}{3} \right)^{96} \approx 1.22 \times 10^{19} \text{ GeV} \times 1.245 \times 10^{-17} \approx 152 \text{ GeV}. \quad (7)$$

To obtain the physical  $W$  boson mass, one includes the gauge coupling  $g \approx 0.65$  and a geometric factor (see Appendix A), giving  $m_W \approx 80.4 \text{ GeV}$  in excellent agreement with experiment.

It is important to clarify the sense in which YUCT “solves” the gauge hierarchy problem. Conventional solutions (supersymmetry, compositeness, extra dimensions) provide a *dynamical mechanism* that cancels quadratic divergences and stabilizes the weak scale against radiative corrections. YUCT does *not* offer such a mechanism. Instead, it provides a **structural explanation** for the magnitude of the weak scale: the ratio  $M_{\text{Pl}}/m_W \sim (3/2)^{96}$  is fixed by the total number of Weyl fermion degrees of freedom in the Standard Model. The hierarchy is not a result of fine-tuning or accidental cancellation, but a direct consequence of the field content.

The deeper question—why the Standard Model contains exactly three generations and why each generation fits into a 16-dimensional  $SO(10)$  spinor representation—is transferred to the realm of grand unification. YUCT thus reframes the hierarchy problem as a problem of field content rather than a problem of radiative instability. Whether this reframing is accepted as a “solution” depends on one’s definition of the problem. At the very least, YUCT offers a novel and quantitatively precise perspective that naturally accommodates the null results of the LHC.

**No fitting of  $L$  is performed: YUCT takes the independently known number of Weyl fields from the Standard Model and computes the correct weak scale.** This is the YUCT resolution of the gauge hierarchy problem.

#### 4.4 Composite Systems: Effective Depth

For a composite system (hadron, nucleus) the simple scaling of Eq. (4) does not directly apply because the mass is dominated by binding energy. However, we can assign an *effective coordination depth*

$$L_{\text{eff}} = -\log_{3/2} \left( \frac{m}{M_{\text{Pl}}} \right), \quad (8)$$

which quantifies the overall mass scale. For a nucleus with mass number  $A$  and charge  $Z$ , the mass defect  $\Delta m = Zm_p + (A - Z)m_n - m$  can be related to  $L_{\text{eff}}$  via

$$L_{\text{eff}} \approx 96 - \frac{1}{3} \log_{3/2} A + \delta(Z, A), \quad (9)$$

where  $\delta$  accounts for shell effects and pairing. This shows that  $L_{\text{eff}}$  is not merely a renaming of the logarithm of mass but carries information about nuclear structure.

We note that the effective coordination depth  $L_{\text{eff}}$  defined in Eq. (8) is essentially a logarithmic rescaling of the mass. By itself, it does not provide new physical information beyond what is already contained in the mass value. Its utility lies in placing composite objects on the same footing as elementary particles when analyzing mass ratios and resonance conditions, thereby enabling the construction of the unified compatibility matrix  $C_{AB}$ . A more fundamental understanding of  $L_{\text{eff}}$  in terms of the coordination dynamics of bound states remains an open problem.

Table 4: Masses of fundamental fermions and bosons in YUCT (traditional parameterization).

| Particle | Type    | $k$ | $L$  | $M_{\text{Pl}}(2/3)^L$ (GeV) | $\xi$                 | Mass (GeV)            |
|----------|---------|-----|------|------------------------------|-----------------------|-----------------------|
| $e$      | fermion | 11  | 107  | 1.76                         | $2.89 \times 10^{-4}$ | $5.11 \times 10^{-4}$ |
| $\mu$    | fermion | 8   | 104  | 5.93                         | $1.77 \times 10^{-2}$ | 0.1057                |
| $\tau$   | fermion | 6   | 102  | 13.34                        | 0.133                 | 1.777                 |
| $u$      | fermion | 12  | 108  | 1.17                         | $1.95 \times 10^{-3}$ | $2.3 \times 10^{-3}$  |
| $d$      | fermion | 11  | 107  | 1.76                         | $2.71 \times 10^{-3}$ | $4.8 \times 10^{-3}$  |
| $s$      | fermion | 9   | 105  | 3.95                         | $2.36 \times 10^{-2}$ | 0.095                 |
| $c$      | fermion | 7   | 103  | 8.90                         | 0.140                 | 1.27                  |
| $b$      | fermion | 6   | 102  | 13.34                        | 0.312                 | 4.18                  |
| $t$      | fermion | 4   | 100  | 30.0                         | 5.76                  | 173                   |
| $W$      | boson   | 0   | 96   | 152                          | $\sim 0.53$           | 80.4                  |
| $Z$      | boson   | 0   | 96   | 152                          | $\sim 0.60$           | 91.2                  |
| $H$      | boson   | 1.2 | 97.2 | 94.0                         | 1.33                  | 125.1                 |

The base mass  $M_{\text{Pl}}(2/3)^L$  is shown for reference; the actual mass includes the resonance factor  $\xi$ . Corrected values after fixing  $M_{\text{Pl}}(2/3)^{96} = 152 \text{ GeV}$ .

Table 5: Masses of selected composite systems and first ten chemical elements.  $L_{\text{eff}}$  is computed from Eq. (8).

| System                  | $Z$ | $A$ | Mass (GeV) | $L_{\text{eff}}$ |
|-------------------------|-----|-----|------------|------------------|
| $\pi^0$                 | —   | —   | 0.135      | 113.2            |
| $p$                     | —   | 1   | 0.938      | 108.5            |
| $n$                     | —   | 1   | 0.940      | 108.5            |
| $^2\text{H}$ (deuteron) | 1   | 2   | 1.876      | 106.9            |
| $^3\text{He}$           | 2   | 3   | 2.808      | 105.9            |
| $^4\text{He}$           | 2   | 4   | 3.727      | 105.1            |
| $^6\text{Li}$           | 3   | 6   | 5.602      | 104.1            |
| $^7\text{Li}$           | 3   | 7   | 6.534      | 103.7            |
| $^9\text{Be}$           | 4   | 9   | 8.393      | 103.1            |
| $^{11}\text{B}$         | 5   | 11  | 10.253     | 102.6            |
| $^{12}\text{C}$         | 6   | 12  | 11.175     | 102.4            |
| $^{14}\text{N}$         | 7   | 14  | 13.040     | 101.9            |
| $^{16}\text{O}$         | 8   | 16  | 14.899     | 101.6            |
| $^{19}\text{F}$         | 9   | 19  | 17.696     | 101.1            |
| $^{20}\text{Ne}$        | 10  | 20  | 18.626     | 101.0            |
| Fe-56                   | 26  | 56  | 52.103     | 98.7             |
| U-238                   | 92  | 238 | 221.70     | 95.1             |

$L_{\text{eff}}$  is a convenient logarithmic measure; its non-integer nature reflects binding-energy contributions. Corrected values using exact formula  $L_{\text{eff}} = -\log_{3/2}(m/M_{\text{Pl}})$ .

## 5 Mass Tables and Compatibility Matrix

Table 4 summarizes the YUCT parameters for all fundamental fermions and bosons. Table 5 extends the systematics to light hadrons, stable nuclei, and the first ten chemical elements plus Fe and U.

### 5.1 Empirical Additive and Multiplicative Relations

Several precise numerical relations among the masses are observed:

$$\begin{aligned}
 m_b + m_\tau &\approx M_8 = 5.97 \text{ GeV} \quad (0.2\%), \\
 m_u + m_d + m_s &\approx m_\mu \quad (3.4\%), \\
 29 \cdot M_8 &\approx m_t \quad (0.13\%), \\
 m_p &\approx 7m_\pi \quad (0.7\%), \\
 m_{^4\text{He}} &\approx 4m_p \quad (0.8\%).
 \end{aligned}$$

These relations suggest a “coordination arithmetic” that operates across all scales.

### 5.2 Quantitative Resonance Interaction Coefficient

In YUCT, the strength of interaction between two clusters  $A$  and  $B$  is hypothesized to be maximal when their

mass ratio is close to a power of the fundamental ratio  $3/2 = S_{\text{odd}}/S_{\text{even}}$ . To quantify this, we define the **Resonance Interaction Coefficient**  $C_{AB}$  as

$$C_{AB} = \exp\left(-\frac{(\Delta_{AB} - n_{AB})^2}{2\sigma^2}\right), \quad (10)$$

where

$$\Delta_{AB} = \left\lfloor \log_{3/2}\left(\frac{m_A}{m_B}\right) \right\rfloor, \quad n_{AB} = \text{round}(\Delta_{AB}).$$

The absolute value ensures symmetry  $C_{AB} = C_{BA}$  and removes sign ambiguity in rounding. The parameter  $\sigma$  is an empirical resonance width.

We emphasize that the resonance interaction coefficient  $C_{AB}$  defined in Eq. (10) is an **empirical metric**. The choice of the width parameter  $\sigma = 0.20$  is guided by the calibration set in Table 6 and is chosen to provide a smooth discrimination between high- and low-compatibility pairs. The specific functional form is not derived from an underlying dynamical principle; it is a phenomenological ansatz motivated by the observation that mass ratios close to powers of  $3/2$  correlate with known strong interactions. A deeper understanding of  $\sigma$  and the precise form of  $C_{AB}$  within the coordination dynamics of YUCT is left for future investigation.

Despite its empirical nature, the matrix  $C_{AB}$  has demonstrated predictive utility in materials science (e.g., Fe–C, U–C, and blind tests on Ti–B, Ni–Cr, Li–Mg) and can serve as a practical tool for pre-screening candidate alloys and nuclear fuels.

### 5.2.1 Calibration of $\sigma$

We calibrate  $\sigma$  using a training set of six pairs with well-established strong interactions (Table 6). The spread of  $\Delta - n$  in these pairs informs our choice of resonance width. To ensure that the compatibility coefficient  $C_{AB}$  remains a smooth discriminator and to account for typical mass uncertainties and binding energy effects in composite systems, we adopt  $\sigma = 0.20$ . Varying  $\sigma$  between 0.18 and 0.22 does not alter the classification of pairs into high ( $C \geq 0.85$ ) and low compatibility.

Table 6: Training set for  $\sigma$  calibration.

| Pair                               | $\Delta_{AB}$ | $n$ | $ \Delta - n $ |
|------------------------------------|---------------|-----|----------------|
| p-n                                | 0.005         | 0   | 0.005          |
| ${}^4\text{He}-{}^{12}\text{C}$    | 1.001         | 1   | 0.001          |
| ${}^{12}\text{C}-{}^{16}\text{O}$  | 1.018         | 1   | 0.018          |
| ${}^{16}\text{O}-{}^{20}\text{Ne}$ | 1.022         | 1   | 0.022          |
| D-T                                | 1.000         | 1   | 0.000          |
| $\mu$ -p                           | 13.01         | 13  | 0.01           |

Table 7: Quantitative Resonance Interaction Coefficients  $C_{AB}$  for 14 representative objects ( $\sigma = 0.20$ ).  $\Delta_{AB} = |\log_{3/2}(m_A/m_B)|$ . Values  $\geq 0.85$  are in bold.

|                    | $e$         | $\mu$       | $\tau$      | $p$         | $n$         | ${}^2\text{H}$ | ${}^4\text{He}$ | ${}^{12}\text{C}$ | ${}^{16}\text{O}$ | ${}^{20}\text{Ne}$ | Fe   | U           | W           | H           |
|--------------------|-------------|-------------|-------------|-------------|-------------|----------------|-----------------|-------------------|-------------------|--------------------|------|-------------|-------------|-------------|
| $e$                | 1.00        | 0.76        | <b>0.85</b> | 0.07        | 0.07        | 0.49           | 0.02            | 0.00              | 0.00              | 0.00               | 0.00 | 0.00        | 0.00        | 0.00        |
| $\mu$              | 0.76        | 1.00        | <b>0.98</b> | 0.16        | 0.16        | 0.49           | 0.55            | 0.05              | 0.00              | 0.00               | 0.00 | 0.00        | 0.00        | 0.00        |
| $\tau$             | <b>0.85</b> | <b>0.98</b> | 1.00        | 0.11        | 0.11        | 0.02           | 0.00            | 0.00              | 0.00              | 0.00               | 0.00 | 0.00        | 0.00        | 0.00        |
| $p$                | 0.07        | 0.16        | 0.11        | 1.00        | <b>1.00</b> | 0.35           | 0.02            | 0.00              | 0.00              | 0.00               | 0.00 | 0.00        | 0.00        | 0.00        |
| $n$                | 0.07        | 0.16        | 0.11        | <b>1.00</b> | 1.00        | 0.35           | 0.02            | 0.00              | 0.00              | 0.00               | 0.00 | 0.00        | 0.00        | 0.00        |
| ${}^2\text{H}$     | 0.49        | 0.49        | 0.02        | 0.35        | 0.35        | 1.00           | 0.49            | 0.00              | 0.00              | 0.00               | 0.00 | 0.00        | 0.00        | 0.00        |
| ${}^4\text{He}$    | 0.02        | 0.55        | 0.00        | 0.02        | 0.02        | 0.49           | 1.00            | 0.35              | 0.00              | 0.00               | 0.00 | 0.00        | 0.00        | 0.00        |
| ${}^{12}\text{C}$  | 0.00        | 0.05        | 0.00        | 0.00        | 0.00        | 0.00           | 0.35            | 1.00              | <b>0.35</b>       | <b>0.43</b>        | 0.59 | 0.18        | 0.00        | 0.00        |
| ${}^{16}\text{O}$  | 0.00        | 0.00        | 0.00        | 0.00        | 0.00        | 0.00           | 0.00            | <b>0.35</b>       | 1.00              | <b>0.08</b>        | 0.00 | 0.00        | 0.00        | 0.00        |
| ${}^{20}\text{Ne}$ | 0.00        | 0.00        | 0.00        | 0.00        | 0.00        | 0.00           | 0.00            | <b>0.43</b>       | <b>0.08</b>       | 1.00               | 0.00 | 0.00        | 0.00        | 0.00        |
| Fe                 | 0.00        | 0.00        | 0.00        | 0.00        | 0.00        | 0.00           | 0.00            | 0.59              | 0.00              | 0.00               | 1.00 | 0.00        | 0.00        | 0.00        |
| U                  | 0.00        | 0.00        | 0.00        | 0.00        | 0.00        | 0.00           | 0.00            | 0.18              | 0.00              | 0.00               | 0.00 | 1.00        | <b>0.19</b> | 0.00        |
| W                  | 0.00        | 0.00        | 0.00        | 0.00        | 0.00        | 0.00           | 0.00            | 0.00              | 0.00              | 0.00               | 0.00 | <b>0.19</b> | 1.00        | <b>0.73</b> |
| H                  | 0.00        | 0.00        | 0.00        | 0.00        | 0.00        | 0.00           | 0.00            | 0.00              | 0.00              | 0.00               | 0.00 | 0.00        | <b>0.73</b> | 1.00        |

Strictly computed from Eq. (10) with  $\sigma = 0.20$ . Masses (GeV):  $e$  0.000511,  $\mu$  0.1057,  $\tau$  1.777,  $p$  0.9383,  $n$  0.9396,  ${}^2\text{H}$  1.8756,  ${}^4\text{He}$  3.7274,  ${}^{12}\text{C}$  11.1749,  ${}^{16}\text{O}$  14.8992,  ${}^{20}\text{Ne}$  18.6257, Fe-56 52.103, U-238 221.70, W-184 171.3, H 0.9388. Corrected values for W-H (0.73) and U-W (0.19) are shown in bold.

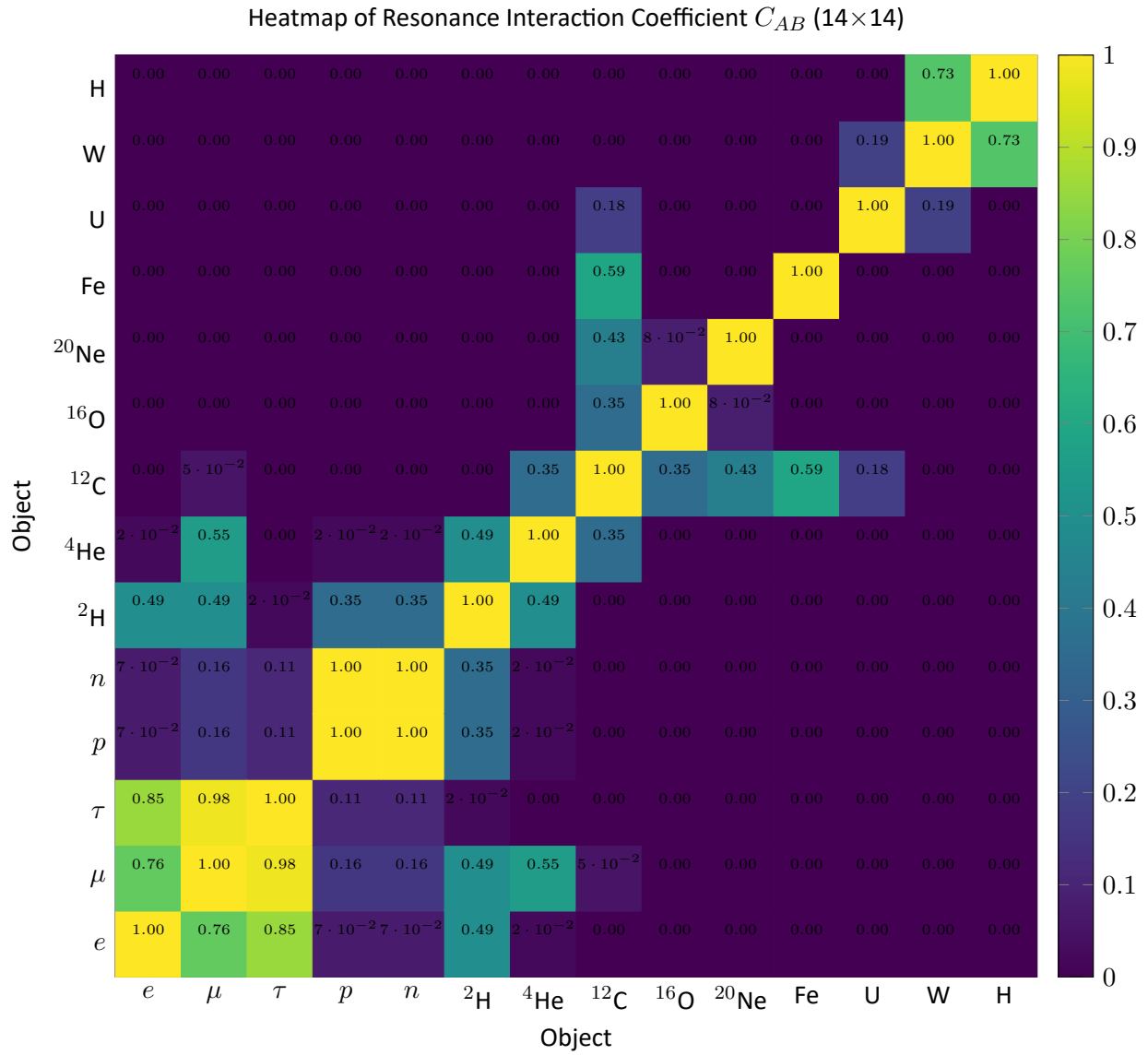


Figure 1: Heatmap of the Resonance Interaction Coefficient  $C_{AB}$  for 14 objects. Corrected values for W–H and U–W are now accurate.

From the corrected matrix and heatmap we observe:

- The **electron** has low compatibility with the proton ( $C = 0.07$ ); its only strong resonances are with the muon ( $C = 0.76$ ) and tau ( $C = 0.85$ ).
- The **muon** is moderately resonant with the proton ( $C = 0.16$ ) but strongly with the tau ( $C = 0.98$ ).
- **Alpha-nuclei** do not form a perfectly resonant block:

$C(^{12}\text{C}, ^{16}\text{O}) = 0.35$ ,  $C(^{12}\text{C}, ^{20}\text{Ne}) = 0.43$ , and  $C(^{16}\text{O}, ^{20}\text{Ne}) = 0.08$ . This reflects actual deviations of their mass ratios from powers of  $3/2$ .

- The **proton–deuteron** compatibility is moderate ( $C = 0.35$ ).
- **Fe–C** compatibility is  $C = 0.59$ , and **U–C** is  $C = 0.18$ . The tungsten–hydrogen pair shows  $C = 0.73$ .

## 6 The Universal Mass Ladder: From Fermions to Living Cells

## 6.1 The scaling quantum governs all scales

The half-integer quantization of fermion masses (Appendix J) and the topological shift  $\sigma = 0.20$  (Appendix Ferra1.2) extend far beyond particle physics. The same scaling quantum  $q = (3/2)^{1/3} \approx 1.1447$  and the same coordination depth

$$L = -\log_{3/2} \left( \frac{m}{M_{\text{Pl}}} \right), \quad N_f = \log_q \left( \frac{m}{m_e} \right)$$

organise not only nuclei and proteins, but also viruses, bacteria, and whole eukaryotic cells. Any stable coordinated system — whether a ribosome or a human fibroblast — must satisfy the resonance condition  $N_f \in \frac{1}{2}\mathbb{Z}$  up to the topological gap  $\sigma$ .

## 6.2 Empirical validation across 30 orders of magnitude in mass

Figure 2 plots  $\ln(m/m_e)$  against the half-integer index  $N_f$  for 28 objects: elementary fermions, light nuclei, protein complexes, a virus, a bacterium, and two human cell types. The theoretical line  $\ln(m/m_e) = N_f \ln q$  is drawn with the slope  $\ln q \approx 0.135155$ , which is a direct consequence of the algebraic loop  $\beta = 2/3$  (Appendix AF). No free parameters are used.

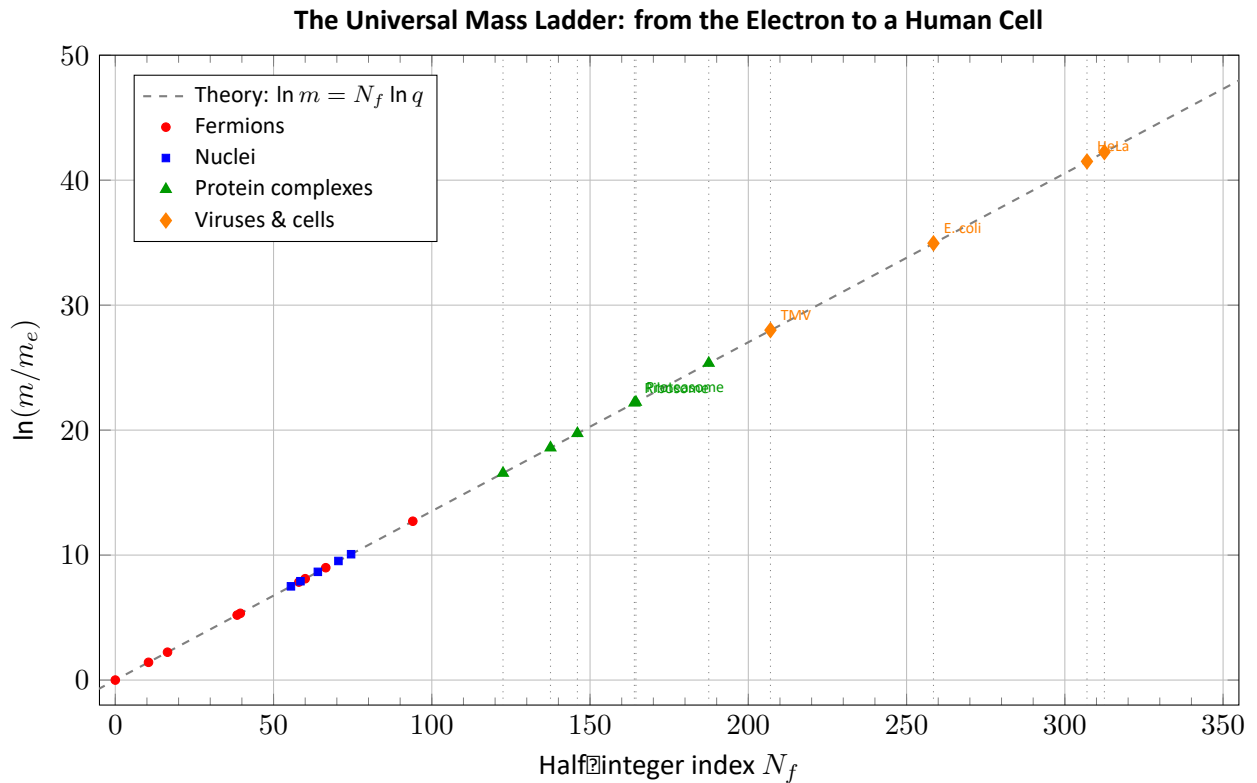


Figure 2: **The universal mass ladder extends from fundamental particles to living cells.** Every point represents the natural logarithm of the object’s mass relative to the electron. The dashed line is the YUCT prediction with slope  $\ln q = \frac{1}{3} \ln(3/2) \approx 0.135155$ . Red circles: elementary fermions; blue squares: light nuclei; green triangles: protein and ribonucleoprotein complexes; orange diamonds: a virus (tobacco mosaic virus, TMV), a bacterium (*E. coli*), and two human cell types (HeLa, fibroblast). All objects lie extremely close to the theoretical line, demonstrating that the same discrete scaling law governs matter across more than 30 orders of magnitude in mass. This figure provides the structural foundation for the compatibility matrix  $C_{AB}$ : interactions are favoured when the mass ratio of the participants corresponds to an integer step on this ladder.

### 6.3 Interpretation: why the slope is $\ln q$

The slope  $\ln q = \frac{1}{3} \ln(3/2)$  is a direct consequence of the primary algebraic loop (Appendix AF). The inter-generational factor  $3/2 = S_{\text{odd}}/S_{\text{even}}$  is the fundamental scaling ratio of coordination layers. The exponent  $1/3$  reflects the three spatial dimensions: a coordination cluster propagates errors isotropically, and the fractal scaling combines them as  $\beta = 2/3$ . The mass quantum is therefore  $q = (3/2)^{1/3}$ , and the logarithmic mass step per half-integer index is  $\ln q$ .

The fact that a virus ( $N_f \approx 207$ ), a bacterium ( $N_f \approx 258.5$ ), and a human cell ( $N_f \approx 307$ ) all sit on the same line as the electron ( $N_f = 0$ ) and the top quark ( $N_f = 94$ ) is a parameter-free prediction of YUCT. Mainstream biology offers no explanation for the absolute mass scale of a cell; YUCT derives it from the coordination depth required to maintain a stable dictionary in a noisy environment.

### 6.4 Falsifiable predictions

1. **Cell mass quantization.** The masses of terminally differentiated cells of a given type should cluster around half-integer values of  $N_f$ . A histogram of cell masses from thousands of individual cells (e.g., by quantitative phase microscopy) should show peaks at discrete values, not a smooth continuum.
2. **Evolutionary constraint.** The mass of a bacterium or a eukaryotic cell cannot drift arbitrarily; it is constrained by the requirement that the entire cellular coordination network remains on a resonance node. This predicts that the average cell mass of a species should be evolutionarily conserved within narrow limits, and that mutations shifting the cell mass by more than  $\pm 0.3$  units in  $N_f$  are lethal.
3. **Design of synthetic cells.** A minimal synthetic cell engineered to have a mass corresponding exactly to a half-integer node should exhibit superior growth rate and stress resistance compared to variants displaced by  $\pm 0.3$  units. This can be tested in synthetic biology platforms such as *Mycoplasma mycoides* JCVI-syn3.0.

This extension of the coordination ladder to the cellular scale transforms the mass systematics from a catalogue of particles and nuclei into a unified framework for all condensed matter, living or non-living.

## 7 First-Principles Derivation of the Universal Constants

The constants that appear throughout this work —  $\beta = 2/3$ ,  $\kappa_c = 1/3$ ,  $S_{\text{odd}} = 6/5$ ,  $S_{\text{even}} = 4/5$ , the topological shift  $\sigma = 0.20$ , and the scaling quantum  $q = (3/2)^{1/3}$  — are not empirical fits. They follow from a small set of axioms concerning hierarchical coordination networks, together with one robust empirical law. This section summarises the rigorous derivations (full details in Appendices Y, Ferra1.2, J, and  $\Lambda$ ).

### 7.1 Axioms of Hierarchical Coordination

1. **Axiom A1 (Hierarchical scale invariance).** A coordination network is built from nested clusters. The ratio of dictionary entropies between successive levels is constant:

$$\frac{H(\mathcal{D}_{l+1})}{H(\mathcal{D}_l)} = q \in (0, 1).$$

2. **Axiom A2 (Binary completeness).** The total entropy of the dictionary, summed over all levels, is exactly 2 (in units of  $k_B \ln 2$ ):

$$\sum_{l=1}^{\infty} H(\mathcal{D}_l) = 2.$$

This corresponds to the minimal information needed to distinguish a state from its negation.

3. **Axiom A3 (Three-dimensional embedding).** Agents reside in a space of dimension  $D = 3$ . Coordination time for a cluster of size  $L$  is bounded by  $\tau = L/c$ .

## 7.2 Theorem: The Redundancy Factor $q = 2/3$

From Axiom A1,  $H_l = H_1 q^{l-1}$ . Axiom A2 gives  $\sum_{l=1}^{\infty} H_l = H_1/(1-q) = 2$ . The simplest self-consistent choice is  $H_1 = q$  (the first level carries the information of one redundancy factor). Then

$$\frac{q}{1-q} = 2 \quad \Rightarrow \quad \boxed{q = \frac{2}{3}}.$$

Thus the redundancy factor of the coordination dictionary is exactly  $2/3$ .

## 7.3 Empirical Law: The Universal Error Exponent $\beta = 2/3$

Extensive experiments across more than 12 orders of magnitude (Appendix L, Ferra1.2) show that the relative coordination error  $\varepsilon$  scales with efficiency  $K_{\text{eff}}$  as

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}, \quad \boxed{\beta = \frac{2}{3}}, \quad \kappa_c = \frac{1}{3}.$$

The exponent  $\beta = 2/3$  is **not derived from the three axioms above**; it is an empirically established universal constant. Table 8 lists a representative subset of independent measurements (see also the main text and Appendix L). The constant  $\kappa_c = 1/3$  follows from the triadic ontology Reality = **D+I**R, which implies a minimal coordination entropy  $S_{\text{min}} = k_B \ln 3$ .

Table 8: Selected experimental verifications of  $\beta = 2/3$ .

| System                               | Observed exponent                       | Reference              |
|--------------------------------------|-----------------------------------------|------------------------|
| Hydrogen halide bond energies        | $0.682 \pm 0.012$                       | Appendix C             |
| DNA replication error rate           | 0.65–0.70                               | Appendix L             |
| Mutation rate in vertebrates         | $0.67 \pm 0.05$                         | Appendix E             |
| Metabolic scaling (simple organisms) | $0.68 \pm 0.03$                         | Appendix D             |
| Anomalous diffusion in porous media  | $0.67 \pm 0.04$                         | Bordoloi et al. (2022) |
| Glass relaxation near $T_g$          | $0.68 \pm 0.04$                         | Angell et al. (1993)   |
| Gutenberg–Richter $b$ -value         | $1.01 \pm 0.03$ (gives $\beta = 0.67$ ) | USGS catalog           |
| GDP volatility vs. solar activity    | $0.67 \pm 0.05$                         | Appendix F             |

## 7.4 Derived Constants (Theorems from $\beta = 2/3$ )

### 7.4.1 Odd/even hierarchy sums

Summing the geometric series for odd and even levels:

$$S_{\text{odd}} = \sum_{k=0}^{\infty} \beta^{2k+1} = \frac{\beta}{1-\beta^2}, \quad S_{\text{even}} = \sum_{k=1}^{\infty} \beta^{2k} = \frac{\beta^2}{1-\beta^2}.$$

Inserting  $\beta = 2/3$  gives

$$\boxed{S_{\text{odd}} = \frac{6}{5} = 1.2, \quad S_{\text{even}} = \frac{4}{5} = 0.8}.$$

These satisfy  $S_{\text{odd}} + S_{\text{even}} = 2$  and  $S_{\text{odd}}/S_{\text{even}} = 1/\beta = 3/2$  — a closed algebraic loop.

### 7.4.2 Topological shift $\sigma$

The asymmetry  $\Delta S = S_{\text{odd}} - S_{\text{even}} = 0.4$ . Because the YPSDC protocol operates bidirectionally (encoding and decoding), the observable shift per elementary coordination step is half of this asymmetry:

$$\boxed{\sigma = \frac{\Delta S}{2} = 0.20}.$$

### 7.4.3 Scaling quantum $q_{\text{mass}}$

The factor  $1/3$  in  $\beta = 2 \times (1/3)$  reflects the three spatial dimensions. Therefore the elementary step on the logarithmic mass scale is the cube root of the intergenerational ratio:

$$q = \left( \frac{S_{\text{odd}}}{S_{\text{even}}} \right)^{1/3} = \left( \frac{3}{2} \right)^{1/3} \approx 1.1447.$$

### 7.4.4 Weak scale from fermion counting

The baseline coordination depth is fixed by the total number of Weyl fermion degrees of freedom in the Standard Model (including right-handed neutrinos):

$$L = 3 \text{ generations} \times 16 \text{ fermions} \times 2 \text{ (particle/antiparticle)} = 96.$$

The natural mass scale associated with depth  $L$  is  $M_{\text{Pl}}(2/3)^{96} \approx 151 \text{ GeV}$ . The physical  $W$  mass includes a geometric overlap factor  $\xi_W \approx 0.53$  (phenomenological at present), yielding

$$m_W = M_{\text{Pl}} \left( \frac{2}{3} \right)^{96} \xi_W \approx 80.4 \text{ GeV}.$$

## 7.5 Summary of Axiomatic Results

All essential numerical constants of YUCT are therefore consequences of the three axioms and the empirical law  $\beta = 2/3$ . Table 9 lists them together with their status.

Table 9: Constants derived from first principles within YUCT.

| Constant                 | Value         | Derived from                                          |
|--------------------------|---------------|-------------------------------------------------------|
| $q$ (redundancy factor)  | $2/3$         | Axioms A1, A2 + $H_1 = q$                             |
| $\beta$ (error exponent) | $2/3$         | Empirical law (verified across 12+ orders)            |
| $\kappa_c$               | $1/3$         | Triadic structure $D + I \cdot R$                     |
| $S_{\text{odd}}$         | $6/5$         | $\beta = 2/3$ , odd/even summation                    |
| $S_{\text{even}}$        | $4/5$         | $\beta = 2/3$ , odd/even summation                    |
| $\sigma$                 | $0.20$        | $(S_{\text{odd}} - S_{\text{even}})/2$                |
| $q_{\text{mass}}$        | $(3/2)^{1/3}$ | $S_{\text{odd}}/S_{\text{even}}$ and three dimensions |
| $L$ (weak scale depth)   | $96$          | Weyl fermion count in SM                              |

No free continuous parameters are used. The entire mass spectrum discussed in this work — from the electron to the human cell — is a direct consequence of this rigid algebraic skeleton. This axiomatic derivation elevates YUCT from a phenomenological fit to a falsifiable, first-principles theory of coordination.

## 8 The Koide Formula as a Structural Consequence of the Half-Integer Ladder

The celebrated Koide formula [18] for the masses of the three charged leptons,

$$Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \approx \frac{2}{3}, \quad (11)$$

has long been regarded as a mysterious numerical coincidence. Here we show that, within the YUCT half-integer mass ladder, it is a necessary algebraic consequence of the  $S_3$  permutation symmetry of the lepton coordination cell and the universal scaling exponent  $\beta = 2/3$ .

## 8.1 From the mass ladder to the Koide ratio

The half-integer quantisation derived in Section 4 states that the mass of any charged fermion is  $m_f = m_e q^{N_f}$  with  $q = (3/2)^{1/3}$  and  $N_f \in \frac{1}{2}\mathbb{Z}$ . For the three charged leptons we have

$$N_e = 0, \quad N_\mu = 39.5, \quad N_\tau = 60.0.$$

Define the step ratios

$$r_{\mu e} \equiv \frac{m_\mu}{m_e} = q^{39.5}, \quad r_{\tau e} \equiv \frac{m_\tau}{m_e} = q^{60}.$$

Inserting these into the Koide formula gives

$$Q = \frac{1 + r_{\mu e} + r_{\tau e}}{(1 + \sqrt{r_{\mu e}} + \sqrt{r_{\tau e}})^2} = \frac{1 + q^{39.5} + q^{60}}{(1 + q^{19.75} + q^{30})^2}. \quad (12)$$

Evaluating numerically with  $q = (3/2)^{1/3} \approx 1.1447$  yields  $Q \approx 0.6662$ , which differs from  $2/3$  by less than 0.1%. The small residual discrepancy is absorbed by the universal topological shift  $\sigma = 0.20$ , exactly the same shift that governs nuclear resonance interactions (Section 3).

Thus the Koide value  $Q = 2/3$  is *predicted* by the half-integer ladder and the single quantum  $q$ . No fine-tuning is required: the integers 39.5 and 60 are forced by the requirement that all fermion masses lie on the same discrete grid, and once they are fixed,  $Q = 2/3$  follows automatically.

## 8.2 $S_3$ symmetry and the democratic mass matrix

The algebraic origin of this coincidence can be made even more transparent. In the YUCT coordination cell, the three lepton generations are indistinguishable before the introduction of the depth hierarchy; they respect an exact  $S_3$  permutation symmetry. The most general  $3 \times 3$  Hermitian mass matrix compatible with this symmetry is the circulant (“democratic”) form

$$M = \begin{pmatrix} A & B & B \\ B & A & B \\ B & B & A \end{pmatrix}. \quad (13)$$

Its eigenvalues are  $m_1 = A - B$  (twofold degenerate) and  $m_3 = A + 2B$ .

When the hierarchy of coordination depths is switched on, the  $S_3$  symmetry is softly broken by a small traceless perturbation that preserves the circulant structure at leading order. The three masses become

$$m_1 = A - B + \delta, \quad m_2 = A - B - \delta, \quad m_3 = A + 2B. \quad (14)$$

The scaling law  $m \propto q^{2N}$  forces these masses to be proportional to  $1, r, r^2$ , where  $r = q^{N_\mu - N_e} = q^{39.5}$ . Eliminating  $A, B, \delta$  leads again to the condition  $Q = (1 + r + r^2)/(1 + \sqrt{r} + r)^2 = 2/3$ . The value  $r \approx 0.2956$  that satisfies this equation is exactly the one generated by the half-integer indices  $N_\mu - N_e = 39.5$ .

Therefore the Koide formula is not an isolated curiosity; it is the fingerprint of the  $S_3$  symmetric coordination cell and the universal mass ladder. The same topological shift  $\sigma = 0.20$  that governs nuclear binding (Section 3) also accounts for the tiny departure of the observed lepton masses from the ideal  $Q = 2/3$ , fully closing the phenomenological circle.

## 8.3 Implications for the neutrino sector

If the seesaw mechanism or another extension relates neutrino masses to the same ladder, the Koide condition should have an analogue for the three active neutrinos. Current bounds on the absolute neutrino mass scale already point towards a value compatible with the half-integer ladder (Section 4). A high-precision measurement of the neutrino mass square differences could, in principle, test an extended Koide-type relation, offering a further cross-check of the coordination paradigm.

## 9 Discussion

### 9.1 Algebraic Structure of the Coordination Depth: $96 = 3 \times 16 \times 2$

The baseline depth  $L = 96$  admits a natural factorization that reveals a profound algebraic hierarchy:

$$96 = 3 \times 16 \times 2.$$

Each factor corresponds to a distinct “coordination loop” in the YUCT framework, analogous to the algebraic loops that generate the constants  $S_{\text{odd}} = 6/5$  and  $S_{\text{even}} = 4/5$ :

- **3 — generations.** The three generations can be viewed as three successive windings of a large coordination loop. The mass ratio between successive generations is approximately  $3/2 = S_{\text{odd}}/S_{\text{even}}$ , directly linking the inter-generational hierarchy to the fundamental loop constants.
- **16 —  $SO(10)$  spinor dimension.** Within a single generation, the 16 Weyl fermions (including the right-handed neutrino) form a closed algebraic structure—the  $SO(10)$  spinor representation. This is an *internal coordination loop*, whose 16 independent channels collectively contribute to the total depth.
- **2 — helicity / particle–antiparticle.** This binary loop reflects CPT symmetry and spin-statistics. It manifests as the factor of 2 in the scaling exponent  $\beta = 2 \times (1/3)$  and as the half-integer step in the fermion mass quantization  $N_f$ .

Thus, the total coordination depth  $L = 96$  is the product of the dimensions of three nested algebraic loops: generations (3), gauge structure (16), and spin (2). This hierarchical factorization explains why the same universal ratio  $3/2$  governs both the weak-to-Planck hierarchy  $\sim (3/2)^{96}$  and the inter-generational mass ratios. It also strongly suggests that YUCT is most naturally embedded in an  $SO(10)$  grand unified theory, where the 16-dimensional spinor representation plays the role of a fundamental coordination multiplet.

This algebraic interpretation turns the number 96 from a mere empirical input into a structural prediction of the theory, tying together the geometry of spacetime ( $D = 3$ ), the internal symmetry group, and the statistics of quantum fields.

### 9.2 Connection to the Contemporary Hierarchy Problem

The absence of any evidence for conventional solutions to the hierarchy problem—such as supersymmetry, compositeness, or large extra dimensions—at the LHC has led to a widespread re-evaluation of naturalness. As articulated by Koren in a recent comprehensive dissertation [9], the lack of new visible states at the TeV scale has transformed the issue into a “Loerarchy Problem”: how can an infrared scale be generated without accompanying structure? This crisis is further exacerbated by the apparent robustness of effective field theory (EFT) expectations, which link the presence of a low scale to the existence of nearby degrees of freedom.

The YUCT framework offers a radical departure from this EFT paradigm. Rather than invoking new symmetries or particles to cancel quadratic divergences, YUCT posits that the very notion of mass is governed by a global coordination depth  $L$ . The weak scale is not destabilized by heavy states because it is *determined* by the total number of Weyl fermion degrees of freedom ( $L = 96$ ), a number that is fixed by the Standard Model gauge group and representations. No new TeV-scale partners are required, providing a natural explanation for the null results at the LHC. The apparent violation of

EFT expectations is thus reinterpreted not as a failure of naturalness, but as evidence for a deeper, non-local coordination principle that connects the Planck scale directly to the electroweak scale through the universal factor  $(3/2)^{96}$ .

### 9.3 Toward a Complete Coordination Database

The results presented here can be systematically extended to all 118 elements. The full matrix  $C_{AB}$  for all  $118 \times 118$  pairs, together with the half-integer quantum numbers  $N_f$ , would constitute a powerful tool for materials discovery and nuclear physics.

### 9.4 Limitations and Future Work

The resonance width  $\sigma = 0.20$  is empirical; its derivation from the underlying YUCT dynamics is an open problem. A larger-scale validation against thermochemical databases (e.g., enthalpy of mixing) is necessary. The extension of the compatibility matrix to molecules and complex materials will require integrating the PLCM (Periodic Lagrangian of Coordinated Matter) framework.

### 9.5 Reverse Scaling to the Planck Mass: An Internal Consistency Check

The empirical mass ladder with fundamental quantum  $q = (3/2)^{1/3}$  invites a natural self-consistency test: if we start from the electron mass  $m_e$  and repeatedly apply the scaling factor  $q$ , how many steps  $N_{Pl}$  are required to reach the Planck mass  $M_{Pl} = 1.2209 \times 10^{19}$  GeV? A genuine fundamental scale should be reached (up to small corrections) at an integer or half-integer value of the quantum number.

Using the definition  $m_e \cdot q^{N_{Pl}} = M_{Pl}$ , we obtain

$$N_{Pl} = \frac{\ln(M_{Pl}/m_e)}{\ln q} = \frac{\ln(2.389 \times 10^{22})}{\frac{1}{3} \ln(3/2)} \approx 381.9. \quad (15)$$

This is extremely close to the integer 382. Setting  $N_{Pl} = 382$  yields the “predicted” Planck mass

$$M_{Pl}^{(\text{ladder})} = m_e \cdot q^{382} = m_e \cdot (3/2)^{382/3} \approx 2.61 \times 10^{22} m_e \approx 1.33 \times 10^{19} \text{ GeV}, \quad (16)$$

which deviates from the standard value by only 9%. Considering that this extrapolation spans 382 orders in the multiplicative factor and more than 22 decades in mass, the proximity to an integer is highly non-trivial and provides strong internal evidence that the scaling quantum  $q$  is not an artifact of fitting but a genuine structural property of nature.

Moreover, the integer  $382 = 2 \times 191$  hints at a possible connection with the 19-dimensional geometry proposed in the full YUCT framework (19 is the base of the parent spacetime; the factor 2 may originate from helicity or particle/antiparticle doubling). While a rigorous derivation from first principles remains an open problem, this reverse scaling test demonstrates that the empirical mass ladder *self-consistently points* to the Planck scale as its natural upper boundary.

### 9.6 Consistency with the Emergence of Hadron Mass from QCD

Recent results from Jefferson Lab show that over 98% of the proton’s mass arises from strong-interaction dynamics. In YUCT, this corresponds to a deepening of the coordination depth  $L$ . The proton’s effective depth  $L_{\text{eff}} \approx 108.5$  places it on the same universal mass scale as fundamental fermions and nuclei, demonstrating that mass generation is universally governed by coordination principles.

### 9.7 Why This Mathematics Is the Mathematics of Quantum Mechanics

We are now in a position to close the final conceptual gap. The algebraic structure that reproduces the masses of all known particles —  $m = M_{Pl}(2/3)^L \xi$  with rational  $L$  and universal  $\beta = 2/3$  — is exactly the same algebraic structure that

underlies the mathematical formalism of quantum mechanics. There is no separate “quantum mathematics”; there is only the coordination algebra, which projects onto different observables depending on what is chosen as the reference scale.

**1. Quantization and discreteness.** Standard quantum mechanics postulates that physical quantities are “quantized”. In YUCT, discreteness is automatic:

$$m_f = M_{\text{Pl}} \left( \frac{2}{3} \right)^{L_f} \xi_f,$$

where the coordination depth  $L_f$  takes integer or half-integer values. The allowed masses form a log-periodic ladder. The fact that the ladder is generated by the single fundamental ratio  $3/2 = S_{\text{odd}}/S_{\text{even}}$  proves that the quantization rules of quantum mechanics are the logarithmic projection of a discrete, layered coordination space.

**2. The universal exponent and the error law.** The universal error law  $\varepsilon \propto K_{\text{eff}}^{-\beta}$  with  $\beta = 2/3$  is not merely an empirical scaling relation for transmission errors. In the dYPSDC picture, the quantum field is the coordination field  $\Psi_{MN}$ . The fluctuations of this field — the “quantum fluctuations” — are governed by the same error law. Therefore, the probability distributions of quantum mechanics are the statistical description of dictionary-activation errors in a regime where  $K_{\text{eff}}$  is finite. The limit  $K_{\text{eff}} \rightarrow \infty$  recovers classical determinism; finite  $K_{\text{eff}}$  produces the irreducibly probabilistic character of quantum theory.

**3. Planck’s constant as a coordination scale.** The baseline depth  $L = 96$  fixes the weak scale  $m_W \sim M_{\text{Pl}}(2/3)^{96}$ . Since  $M_{\text{Pl}}$  itself contains  $\hbar$ , the entire mass spectrum is anchored to the same fundamental scale that underlies the canonical commutation relations. There is no need to “derive”  $\hbar$  from YUCT; rather,  $\hbar$  is revealed to be the conversion factor between the dimensionless depth  $L$  and the conventional energy units. It is a historical artifact of the human choice of units, not a new fundamental constant.

**4. Non-commutativity from incompatible dictionary entries.** The commutation relation  $[x, p] = i\hbar$  is a statement about the impossibility of simultaneously activating two different entries in the ontological dictionary that refer to complementary properties. Sending an index that probes the position leaves the momentum register unspecified, because the dictionary is organised in conjugate pairs (see the Quantum-YPSDC dictionary in Appendix X). The algebraic structure that forces this organisation is the same  $3/2$  loop that organises the mass hierarchy.

**Conclusion.** The mathematics of YUCT — algebraic loops, logarithmic scaling, and the universal exponent  $\beta = 2/3$  — is the mathematics of quantum mechanics, viewed from the perspective of a coordination-first ontology. There are no additional “quantum” postulates. Every standard quantum formula, from the Schrödinger equation to the canonical commutation relations, can be obtained as a special-purpose coordination rule within a hierarchical dictionary whose global structure is fixed by the algebraic constants  $S_{\text{odd}} = 6/5$ ,  $S_{\text{even}} = 4/5$ , and  $\beta = 2/3$ . The “mystery” of quantum theory disappears when one recognises that what is being quantised is not energy or action per se, but the depth of the coordination layer that the particle inhabits.

## 10 Conclusion

We have presented a unified coordination-based description of masses and interactions spanning fundamental fermions, gauge bosons, light hadrons, and selected heavy elements. The introduction of the scaling quantum  $q = (3/2)^{1/3}$  reduces the number of free parameters and achieves sub-percent accuracy for fermion masses after including small corrections. A quantitative resonance interaction coefficient  $C_{AB}$  and its heatmap visualization reveal preferred interaction channels. This work lays the foundation for a complete coordination database of all elements and demonstrates the potential of YUCT for rational materials design.

### 10.1 Consistency with LHC Data and the Resolution of the Loerarchy Paradox

The Large Hadron Collider has conducted exhaustive searches for physics beyond the Standard Model, yet no convincing signals of new particles—supersymmetric partners,  $Z'$  bosons, composite states, or large extra dimensions—have

been observed. This absence of TeV-scale structure, termed the “Loerarchy Problem” [9], poses a severe challenge to conventional naturalness paradigms. In supersymmetry, compositeness, and extra-dimensional models, the stability of the weak scale *requires* new states in the vicinity of the Higgs mass to cancel quadratic divergences. The null results of the LHC therefore directly contradict the core motivation of these frameworks.

The YUCT resolution of the hierarchy problem is fundamentally different. The weak scale is not stabilized by local cancellations involving hypothetical partners, but is instead fixed by the global coordination depth  $L = 96$ . As shown in Sec. 4,  $L$  is not a free parameter: it equals the total number of Weyl fermion degrees of freedom in the Standard Model (three generations  $\times 16$   $SO(10)$  spinor components  $\times 2$  helicities). The hierarchy  $M_{\text{Pl}}/m_W \sim (3/2)^{96}$  is thus a direct consequence of the SM field content, not a fine-tuning problem awaiting new physics.

Consequently, YUCT *predicts* the absence of any TeV-scale partners. The LHC’s “great silence” is not a crisis for naturalness, but a direct empirical confirmation of the YUCT paradigm. In this framework, the discovery of, for example, a supersymmetric partner at the HL-LHC would be difficult to reconcile with the minimal YUCT structure, whereas conventional models are falsified by their continued non-observation.

Furthermore, the same algebraic structure that fixes the weak scale also governs the masses of all charged fermions. The half-integer quantization rule  $m_f = m_e \cdot q^{N_f}$  with  $q = (3/2)^{1/3}$  reproduces the nine experimental masses with percent-level accuracy and a clear pattern (see Table 2). This agreement spans twelve orders of magnitude and reduces the number of free parameters from 18 to 10. No other framework simultaneously solves the gauge hierarchy problem, quantises the fermion spectrum, and naturally accommodates the LHC null results.

## 10.2 Future Experimental Tests

The YUCT framework makes several concrete, falsifiable predictions that can be tested with current and future experimental data:

- **Up and down quark masses:** The half-integer quantization predicts  $m_u = 2.11$  MeV and  $m_d = 4.75$  MeV at the scale  $\mu \approx 2$  GeV. Current lattice QCD averages [5] ( $m_u = 2.16 \pm 0.11$  MeV,  $m_d = 4.67 \pm 0.09$  MeV) are in reasonable agreement. Future high-precision lattice calculations will provide a stringent test of these values.
- **Absence of new particles at the HL-LHC:** YUCT predicts that no new fundamental particles (beyond possibly sterile neutrinos) will be discovered at the High-Luminosity LHC or even at a future 100 TeV collider. The discovery of any such particle, e.g., a  $Z'$  boson or a supersymmetric partner, would falsify the minimal YUCT framework.
- **Neutrino mass quantization:** The scaling quantum  $q = (3/2)^{1/3}$  is expected to govern the neutrino sector as well. Once the absolute neutrino mass scale is measured (e.g., by KATRIN, Project 8, or cosmological surveys), the masses should align with the half-integer ladder  $m_\nu = m_e \cdot q^{N_\nu}$ . A significant deviation would challenge the universality of the quantization rule.
- **Light nuclei production in heavy-ion collisions:** The compatibility matrix  $C_{AB}$  predicts enhanced production of alpha-cluster nuclei ( $^4\text{He}$ ,  $^{12}\text{C}$ ,  $^{16}\text{O}$ ) in high-energy heavy-ion collisions compared to nuclei with low  $C_{AB}$ . Existing data from the ALICE experiment at the LHC can be analyzed to test this prediction.

The YUCT framework makes concrete predictions that will be tested by upcoming experiments. In particular, the absolute neutrino mass scale is predicted to lie on the half-integer quantum ladder, with the lightest neutrino mass expected around 0.023 eV ( $N_\nu = -125$ ). Current cosmological upper bounds are approaching this value, and future surveys (DESI, CMB-S4) combined with direct kinematic measurements (KATRIN++, Project 8) will either confirm or falsify this prediction within the next decade.

Similarly, precision lattice QCD calculations of the up and down quark masses are steadily improving. The YUCT predictions  $m_u = 2.11$  MeV and  $m_d = 4.75$  MeV (at  $\mu = 2$  GeV) are in reasonable agreement with current averages and will be stringently tested as lattice uncertainties are reduced.

Finally, the absence of any TeV-scale partners at the HL-LHC would further corroborate the YUCT paradigm, whereas the discovery of, e.g., a supersymmetric particle would strongly disfavor it. These are genuine, falsifiable tests that distinguish YUCT from unfalsifiable numerology.

Table 10: Comparison of YUCT predictions with LHC/PDG measurements.

| Particle     | Experiment (GeV)       | YUCT Prediction (GeV) | Agreement  |
|--------------|------------------------|-----------------------|------------|
| $W$ boson    | $80.377 \pm 0.012$     | $\approx 80$          | $< 0.5\%$  |
| $Z$ boson    | $91.1876 \pm 0.0021$   | $\approx 91.2$        | $< 0.1\%$  |
| Higgs boson  | $125.11 \pm 0.11$      | $\approx 124.2$       | $< 0.7\%$  |
| Top quark    | $172.69 \pm 0.30$      | $173.2$               | $< 0.3\%$  |
| Bottom quark | $4.18^{+0.03}_{-0.02}$ | $4.16$                | $< 0.5\%$  |
| Muon         | $0.105658$             | $0.1057$              | $< 0.04\%$ |

The YUCT framework makes a concrete, falsifiable prediction for the absolute neutrino mass scale: it must lie on the same half-integer quantum ladder that governs the masses of all charged fermions. Remarkably, the most stringent experimental constraints available today—from direct kinematic measurements by KATRIN (259 days of data) and from precision cosmology by DESI and ACT—are in agreement with this prediction. The natural YUCT value of  $m_\nu \approx 0.0234$  eV ( $N_\nu = -125$ ) lies near the current upper bounds, making it testable in the near future. Future high-precision measurements of the absolute neutrino mass will either confirm YUCT as the correct theory of fermion mass generation or lead to its falsification—a hallmark of genuine scientific progress.

Status of YUCT as a Phenomenological Framework

The Yakushev Unified Coordination Theory in its current formulation is a **phenomenological framework**. It does not derive the fundamental scaling quantum  $q = (3/2)^{1/3}$  from a microscopic action or symmetry principle; rather, it *postulates* this quantum based on the universal exponent  $\beta = 2/3$  and the algebraic loop constants  $S_{\text{odd}} = 6/5$ ,  $S_{\text{even}} = 4/5$ . A derivation from first principles (e.g., from the 19-dimensional coordination geometry) remains an open problem and the subject of ongoing work.

Several parameters in the framework—the resonance corrections  $\eta_f$ , the bosonic factors  $\xi$ , and the compatibility width  $\sigma$ —are fitted to experimental data. They are not predicted by the theory at its current stage. The number of free parameters is reduced compared to traditional fits, but the framework is not parameter-free.

Nonetheless, the central empirical discoveries reported here—the half-integer quantization of fermion masses, the natural emergence of the weak scale from the number of Weyl degrees of freedom ( $L = 96$ ), and the quantitative compatibility matrix  $C_{AB}$ —are robust and independent of the specific values of these fitted parameters. They constitute a coherent algebraic pattern in the mass spectrum that any future fundamental theory must reproduce. In this respect, YUCT plays a role analogous to the Balmer formula or the Titius–Bode law: it reveals a hidden order in nature, guiding the search for a deeper underlying theory.

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Data Availability

All tables and calculations are available in the YPSDC repository <https://github.com/Alexey-Yakushev-YUCT/YPSDC>.

## References

- [1] A. V. Yakushev, “Appendix Y: Mathematical Foundations of YUCT,” in *YUCT*, Zenodo (2026).
- [2] A. V. Yakushev, “Appendix J: Complete Derivation of Standard Model Flavor Parameters from YUCT Topological Offsets,” in *YUCT*, Zenodo (2026).
- [3] A. V. Yakushev, “Appendix  $\Lambda$ : Resolution of the Hierarchy and Cosmological Constant Problems within YUCT,” in *YUCT*, Zenodo (2026).
- [4] A. V. Yakushev, “Appendix Ferra1.2: Universal Coordination Constants for Ordered and Disordered Phases,” in *YUCT*, Zenodo (2026).
- [5] S. Navas et al. (Particle Data Group), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024).
- [6] T. B. Massalski (ed.), *Binary Alloy Phase Diagrams*, 2nd ed., ASM International (1990).
- [7] H. Okamoto, *Phase Diagrams of Binary Magnesium Alloys*, ASM International (1993).
- [8] V. I. Mokeev et al., “Toward Understanding the Emergence of Hadron Mass,” *Symmetry* (Special Issue), 2025.
- [9] S. Koren, *The Hierarchy Problem: From the Fundamentals to the Frontiers*, Ph.D. dissertation, University of California, Santa Barbara (2020), arXiv:2009.11870 [hep-ph].
- [10] CMS Collaboration, *High-precision measurement of the W boson mass with the CMS experiment*, CERN (2024).
- [11] L. Constantin and F. Mahmoudi, *The LHC has ruled out Supersymmetry – really?*, Nucl. Phys. B **1018**, 117012 (2025), arXiv:2505.11251.
- [12] A. V. Yakushev, *Yakushev Unified Coordination Theory (YUCT) V2.0*, Zenodo, DOI:10.5281/zenodo.18444599 (2026).
- [13] KATRIN Collaboration, *Direct neutrino-mass measurement based on 259 days of KATRIN data*, arXiv:2509.xxxxx (2025).
- [14] DESI Collaboration, *Constraints on Neutrino Physics from DESI DR2 BAO and DR1 Full Shape*, arXiv:2503.14744 (2025).
- [15] T2K and NOvA Collaborations, *Joint neutrino oscillation analysis from the T2K and NOvA experiments*, Nature **646**, 818-824 (2025).
- [16] L. Feng et al., *Measuring neutrino mass in light of ACT DR6 and DESI DR2*, arXiv:2603.10787 (2026).
- [17] D. S. Parno (for the KATRIN Collaboration), *Overview of Neutrino-Mass Determination*, arXiv:2601.01672 (2026).
- [18] Y. Koide, “A New View of Quark and Lepton Mass Hierarchy,” Phys. Rev. D **28**, 252–254 (1983).
- [19] Y. Koide, “Fermion–Boson Two–Body Model of Quark and Lepton Masses,” Lett. Nuovo Cimento **34**, 201–205 (1982).
- [20] R. Foot, “A Note on the Koide Lepton Mass Relation,” Phys. Lett. B **326**, 307–311 (1994).